

Covered: Symmetric polynomials, elementary symmetric functions, fundamental theorem, lexicographic order, algebraic independence, discriminant, Newton's identities.

1 Symmetric Polynomials

Let R be a commutative ring, and consider $A = R[x_1, \dots, x_n]$. Given $\sigma \in S_n$ and $f = f(x_1, \dots, x_n) \in A$, define

$$\tilde{\sigma}(f) = f(x_{\sigma(1)}, \dots, x_{\sigma(n)}),$$

where $x_i \mapsto x_{\sigma(i)}$. Thus $\tilde{\sigma} : A \xrightarrow{\cong} A$ is an automorphism.

Definition 1.1. $f \in A$ is called *symmetric* if

$$\tilde{\sigma}(f) = f \quad \forall \sigma \in S_n.$$

We denote the set of all symmetric polynomials in A as Σ_n .

Example 1.2.

$$x_1 + x_2 + \dots + x_n, \quad x_1^2 + x_2^2 + x_3^2 + x_1x_2 + x_2x_3 + x_3x_1, \quad x_1x_2 \cdots x_n$$

Exercise 1.3. Σ_n is a subring of A .

Consider $A[T]$, and let

$$E(T) = (T - x_1)(T - x_2) \cdots (T - x_n).$$

Extend $\tilde{\sigma}$ to $A[T]$ by putting $\tilde{\sigma}(T) = T$. Clearly,

$$\tilde{\sigma}(E(T)) = E(T) \quad \forall \sigma \in S_n.$$

We write

$$E(T) = T^n - s_1T^{n-1} + s_2T^{n-2} - \dots + (-1)^n s_n.$$

Then $\tilde{\sigma}(s_i) = s_i$, $\forall \sigma \in S_n$, and hence $s_i \in \Sigma_n$, $1 \leq i \leq n$. We call $s_1, s_2, \dots, s_n$ the symmetric functions of $x_1, x_2, \dots, x_n$.

Comparing coefficients, we have that

$$\begin{aligned} s_1 &= \sum_{i=1}^n x_i, \\ s_2 &= \sum_{1 \leq i < j \leq n} x_i x_j, \\ &\vdots \\ s_n &= x_1 x_2 \cdots x_n. \end{aligned}$$

Theorem 1.4. Every symmetric polynomial in A is a polynomial in $s_1, \dots, s_n$, i.e.

$$\Sigma_n = R[s_1, \dots, s_n].$$

Definition 1.5. A polynomial is called homogeneous if each monomial occurring in it has the same degree, say k .

In particular,

$$\deg(x_1^{k_1} \cdots x_n^{k_n}) = k_1 + \cdots + k_n.$$

Example 1.6.

$$x_1^2 + 9x_1x_2 + 5x_2^2.$$

Remark 1.7. Every symmetric polynomial can be written uniquely as a sum of homogeneous polynomials of different degrees.

Given a monomial $x_1^{k_1} \cdots x_n^{k_n}$ define

$$\deg(x_1^{k_1} \cdots x_n^{k_n}) = k_1 + \cdots + k_n.$$

Also,

$$\deg f = \max\{\deg(\text{monomial occurring in } f)\}.$$

Example 1.8.

$$\begin{aligned} f &= x_1^3 + x_2^3 + x_1x_2 + x_1x_3 + x_2x_3 + x_3^3 \\ &= (x_1^3 + x_2^3 + x_3^3) + (x_1x_2 + x_1x_3 + x_2x_3). \end{aligned}$$

It is sufficient to show that homogeneous symmetric polynomials can be expressed as polynomials in $s_1, \dots, s_n$.

Definition 1.9. Given a monomial $x_1^{k_1} \cdots x_n^{k_n}$ define its multidegree by

$$\text{mdeg}(x_1^{k_1} \cdots x_n^{k_n}) = (k_1, \dots, k_n) \in \mathbb{N}^n.$$

Consider the lexicographic ordering on monomials of degree d :

$$x_1^{k_1} \cdots x_n^{k_n} \succ x_1^{l_1} \cdots x_n^{l_n}$$

if

$$\exists s \geq 0 \text{ such that } k_i = l_i \ (1 \leq i \leq s), \quad k_{s+1} > l_{s+1}.$$

Example 1.10. The following is an example of such a lexicographic ordering,

$$x_1^3x_2x_3 \succ x_1^3x_2^2 \succ x_1x_2^4.$$

Lemma 1.11. Given monomials M_1, M_2, N_1, N_2 with $\deg(M_1) = \deg(M_2) = s$ and $\deg(N_1) = \deg(N_2) = t$ where $s \geq t$. If $M_1 \succeq M_2$ and $N_1 \succeq N_2$ then we have,

$$M_1N_1 \succeq M_2N_2.$$

The above relation follows from

$$M_1 N_1 \succeq M_1 N_2 \succeq M_2 N_2.$$

Remark 1.12. The maximal monomial in s_i is $x_1 x_2 \cdots x_i$. Hence the maximal monomial in $s_1^{d_1} s_2^{d_2} \cdots s_n^{d_n}$ is

$$x_1^{d_1+d_2+\cdots+d_n} x_2^{d_2+\cdots+d_n} \cdots x_n^{d_n}$$

Thus, given $a_1 \geq a_2 \geq \cdots \geq a_n$, the monomial $x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}$ is the maximal monomial in

$$s_1^{a_1-a_2} s_2^{a_2-a_3} \cdots s_{n-1}^{a_{n-1}-a_n} s_n^{a_n}.$$

Proof. Let $x_1^{k_1} \cdots x_n^{k_n}$ be the maximal monomial in f , with coefficient a . Since f is symmetric, if $k_i < k_{i+1}$ for some i , then permuting x_i and x_{i+1} would produce a lexicographically larger monomial occurring in f , contradicting maximality. Hence $k_1 \geq k_2 \geq \cdots \geq k_n$. Put

$$f_1 = f - a s_1^{k_1-k_2} s_2^{k_2-k_3} \cdots s_{n-1}^{k_{n-1}-k_n} s_n^{k_n}.$$

By the above observation, we have,

$$\max \text{mon}(f_1) \prec \max \text{mon}(f).$$

We continue this process. At each step the maximal monomial strictly decreases. Since there are only finitely many monomials of a fixed degree, the process terminates, and hence $f = g(s_1, \dots, s_n)$ for some $g \in R[x_1, \dots, x_n]$. $\square$

Definition 1.13. We say $y_1, \dots, y_n$ are algebraically independent over R if there is no non-zero polynomial $f \in R[x_1, \dots, x_n]$ such that

$$f(y_1, \dots, y_n) = 0.$$

Example 1.14. 1 and e are algebraically independent over R . (also over $\mathbb{Q}$, it is an open problem whether π and e are algebraically independent.)

Remark 1.15. $s_1, \dots, s_n$ are algebraically independent over R .

Proof. For $n = 1$, there is nothing to prove. Suppose, inductively, that $s_1, \dots, s_{n-1}$ are algebraically independent. We then show the case for $s_1, \dots, s_n$. Assume (by way of contradiction) that $s_1, \dots, s_n$ are algebraically dependent. Then,

$$\exists 0 \neq F \in R[x_1, \dots, x_n] \quad \text{such that} \quad F(s_1, \dots, s_n) = 0.$$

We factor the above as,

$$F = F_0(x_1, \dots, x_{n-1}) + F_1(x_1, \dots, x_{n-1})x_n + \cdots + F_d(x_1, \dots, x_{n-1})x_n^d.$$

After factoring out a maximal power of x_n , we may assume that $F_0 \neq 0$. (Otherwise we would have $F = x_n G$ for some G and $\deg G < \deg F$. Since

$0 = F(s_1, \dots, s_n) = s_n G(s_1, \dots, s_n)$ this would contradict the minimality of F). Thus,

$$0 = F_0(s_1, \dots, s_{n-1}) + \dots + F_d(s_1, \dots, s_{n-1})s_n^d$$

Put $x_n = 0$. Then

$$0 = F_0((s_1)_0, \dots, (s_{n-1})_0),$$

where $(s_i)_0$ is the i -th symmetric function in $x_1, \dots, x_{n-1}$. Hence $F_0 \neq 0$ gives a polynomial relation among $s_1, \dots, s_{n-1}$ contradicting the induction hypothesis.

Therefore $s_1, \dots, s_n$ are algebraically independent over R . $\square$

Definition 1.16. The weight of a monomial $x_1^{k_1} \dots x_n^{k_n}$ is

$$w(x_1^{k_1} \dots x_n^{k_n}) = k_1 + 2k_2 + \dots + nk_n.$$

For $f \in R[x_1, \dots, x_n]$, define

$$w(f) = \max\{w(\text{monomial occurring in } f)\}.$$

Theorem 1.17. A symmetric polynomial of degree d can be expressed as $g(s_1, \dots, s_n)$ with $w(g) \leq d$.

Example 1.18. Let $f = x_1^4 + x_2^4 + x_3^4 + \dots$. Then

$$f_1 = f - s_1^4 = -4x_1^3x_2 + \dots$$

$$f_2 = f_1 + 4s_1^2s_2 + \dots$$

$\vdots$

Continuing this process gives the required expression.

2 Discriminant

Let $f(T) = (T - x_1) \dots (T - x_n)$. Define

$$\delta(f) = \prod_{1 \leq i < j \leq n} (x_i - x_j).$$

For $\sigma \in S_n$, $\tilde{\sigma}(\delta(f)^2) = \delta(f)^2$. Hence

$$D(f) := \delta(f)^2 = \prod_{1 \leq i < j \leq n} (x_i - x_j)^2$$

is a symmetric polynomial.

Therefore $D(f) = \Delta(s_1, \dots, s_n)$ for some symmetric polynomial Δ .

Example 2.1. For $n = 2$, $f = x^2 + bx + c$ so,

$$D(f) = b^2 - 4c = s_1^2 - 4s_2.$$

Example 2.2. For $n = 3$, $f = x^3 + ax + b$ and $D(f) = -4a^3 - 27b^2$. Also,

$$D(f) = (x_1 - x_2)^2(x_1 - x_3)^2(x_2 - x_3)^2.$$

3 Newton's Identities

Let $x_1, x_2, \dots, x_n$ be n variables. For $k > 0$, define

$$p_k = \sum_{i=1}^n x_i^k = x_1^k + \dots + x_n^k, \quad p_0 = n$$

Theorem 3.1 (Newton's Identity). For $1 \leq k \leq n$,

$$p_k - s_1 p_{k-1} + s_2 p_{k-2} - \dots + (-1)^{k-1} s_{k-1} p_1 + (-1)^k k s_k = 0.$$

More generally, for each $j \geq 0$,

$$p_{k+j} - s_1 p_{k+j-1} + s_2 p_{k+j-2} - \dots + (-1)^{k-1} s_{k-1} p_{j+1} + (-1)^k s_k p_j = 0$$

Exercise 3.2. Express p_2, p_3, p_4 in terms of the elementary symmetric functions $s_1, s_2, \dots, s_n$.

Exercise 3.3. For $n = 5$, express the following symmetric sum in terms of the elementary symmetric functions,

$$\sum_{1 \leq i < j < k \leq 5} x_i^2 x_j^2 x_k$$